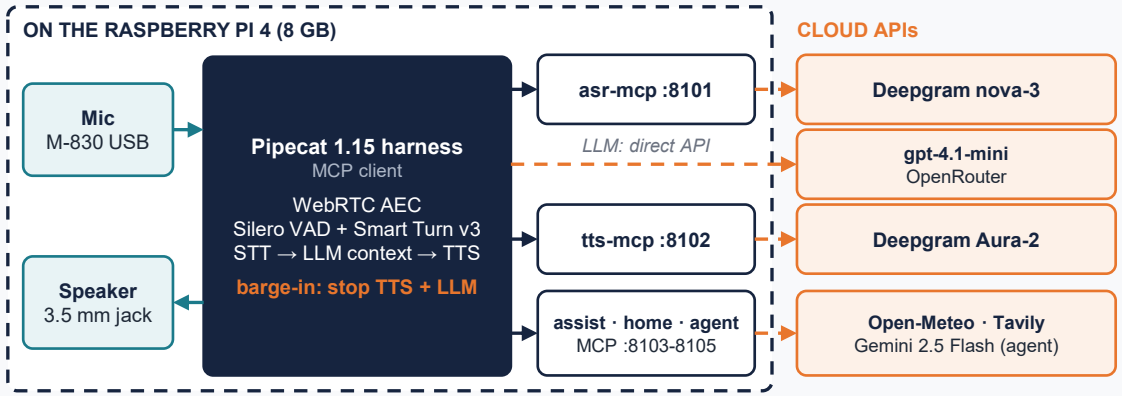
Full-Duplex Voice Agent on a Raspberry Pi

Iris · Pipecat + 5 own MCP servers on a Pi 4

1 SYSTEM DIAGRAM

the one diagram that explains your build




```
mic -> AEC+VAD -> asr-mcp -> LLM(+MCP tools) -> tts-mcp -> speaker
```

```
barge-in: speech detected -> stop TTS + LLM -> listen (0.38 s)
```

- Mic -> WebRTC AEC -> Silero VAD (on RPi) -> asr-mcp -> LLM -> tts-mcp -> speaker
- Harness with MCP client: Pipecat 1.15 (streamable-HTTP MCP; Smart Turn v3 decides end of turn)
- LLM is a commercial API (gpt-4.1-mini via OpenRouter); every tool comes from MY MCP servers
- All 5 MCP servers built by me, on the RPi, started at boot (systemd): asr, tts, assist, home, agent
- Barge-in cancels TTS playback and the LLM stream: playback stops 0.38 s after speech starts
- Real tasks: weather, web search, timers on the TM1637, 3 LED bulbs + OLED, music, background agent builds a web game
- Live web emulator (Firebase): iris-me3-emulator.web.app
- On-RPi vs cloud: dashed box = runs on the Pi; orange = cloud APIs

2 PROMPT

kept in the repo, versioned (assistant/main.py)

SYSTEM PROMPT (v3, PROMPT_VERSION="v3") - excerpt

```
You are Iris, a calm, quick voice agent on a RPi;
  light dry wit, calls the user "boss".
Short, natural sentences; at most two per turn.
If the user speaks, stop and listen; answer what
  they just said, not the old topic.
Confirm before costly or irreversible actions.
Tools via MCP: weather, search, timers, home;
  delegate_task for long jobs. Never read markdown
  or URLs aloud.
```

3 CONTEXT ENGINEERING

GOES IN:

- system prompt v3 + persona + 26 tool schemas (~1.2k tokens)
- ASR final per VAD / Smart Turn segment
- VAD turn events (user-turn aggregation)
- MCP tool results + async agent results
- memory: notes / reminders via tools (no RAG)

MANAGED:

- replies capped at 400 tokens; fillers spoken, kept out of context
- stable prefix (prompt + tools) never edited: cache-friendly TTFT
- music playing: speech must name Iris, so lyrics never enter context
- on barge-in: drop queued TTS audio; echo filter drops self-heard text
- logged per run: **1.2k -> 5.2k tokens** over 30 turns (no trim yet)

MCP Services, Evaluation & Model Serving

build + run servers on the RPi

4 MCP SERVICES

+ section 6: model serving (one table, one truth)

Component	MCP server / transport	Tools	Model	Served where / notes
ASR	asr-mcp · streamable HTTP :8101	transcribe(audio_b64) -> text · warm()	Deepgram nova-3 (+ keyterms)	RPi; final on VAD end; kept warm while user talks
TTS	tts-mcp · HTTP + SSE :8102	synthesize_stream(text) -> 100 ms PCM chunks	Deepgram Aura-2 (pandora)	RPi; chunks as MCP progress; cancel on barge-in
LLM	Pipecat 1.15 (harness MCP client)	function calling: 26 MCP tools + delegate_task	OpenAI gpt-4.1-mini (OpenRouter)	commercial API; streaming tokens; max 400
VAD	in-process (Pipecat)	VADProcessor · Smart Turn v3 · WebRTC AEC	Silero VAD v5 + Smart Turn v3	RPi on-device, 16 kHz; AEC removes own voice
Task	assist-mcp · HTTP :8103	time, weather, web_search, timer, alarm, reminders, notes	Open-Meteo · Tavily	RPi; due timers polled and spoken aloud
Task	home-mcp · HTTP :8104	lights, colour, brightness, thermostat, OLED, timer, music	GPIO LEDs · I2C OLED · VLC	RPi; 3 LED bulbs, OLED, TM1637; music ducks
Task	agent-mcp · HTTP :8105	run_task(task) -> spoken summary + page link	Gemini 2.5 Flash (OpenRouter)	RPi; async background job; pages on :8080

5 EVALUATION & BENCHMARK

standardized latency method (classwork) · King's fdbench

- Method: one Pi clock, every stage timestamped (fdbench); 10 phrases x 3 runs, mean + p95
- Full-Duplex-Bench (40 trials): silent in 2/2 pauses, yielded to interrupt, 2/2 false turns

vs SOTA	Architecture	Response latency	Pauses (FDB)
Iris (ours) · RPi 4	cascade ASR->LLM->TTS + MCP	**4.08 s (median 3.86)**	silent 2/2
GPT-4o voice [1]	native speech-to-speech (cloud)	0.32 s avg (0.23 min)	-
Moshi [2]	full-duplex speech LM (GPU)	0.16 s theory, ~0.2 s real	takes over ~all [3]
ChatGPT Voice '23 [1]	cascade (GPT-4)	5.4 s avg	-

[1] OpenAI GPT-4o launch, 2024 · [2] Kyutai Moshi paper, arXiv 2410.00037 · [3] Full-Duplex-Bench (2503.04721) re-reports. Human turn gap ~0.2 s.

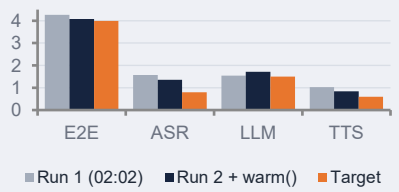
Metric	Definition	Target	Mean	p95
E2E latency	end of speech -> audio out	< 4.0 s	**4.08 s**	5.65 s
ASR final	end of speech -> transcript	< 0.8 s	**1.36 s**	1.58 s
LLM TTFT	transcript -> first token	< 1.5 s	**1.72 s**	3.02 s
TTS TTFB	text -> first audio chunk	< 0.6 s	**0.84 s**	1.05 s
Barge-in	speech -> playback stopped	< 0.8 s	**0.38 s**	0.38 s
WER	ASR accuracy on test set (30)	< 15 %	**6.2 %**	-

DEMO (2-3 min, no push-to-talk)

1 **"Build me a flappy bird game"** -> agent in background
2 **"Lights purple"** -> red + blue bulbs light up
3 **"Weather in Davao?"** -> live get_weather MCP call
4 **"Set a 2-minute timer"** -> countdown on the TM1637
5 **Long answer, I cut in** -> Iris stops in ~0.4 s
6 **"Play music" / "Stop the music"** -> music gate

live view: iris-me3-emulator.web.app

RESULTS (mean s, 30 turns)



Raspberry Pi 4B · 8 GB RAM
heatsink · peak 62 °C · never throttled
Wi-Fi to cloud; 4 ms link to the laptop
2026-11-07 · fdbench 0.1.0
pipecat 1.15 · nova-3 · gpt-4.1-mini
Aura-2 · Silero v5 · Smart Turn v3
E2E median 3.86 s (under target)

CHALLENGES & REFERENCES

1 **Slow ASR/TTS after idle** -> warm() pings **-> ASR 1.57->1.36 s, TTS 1.02->0.84 s**
2 **TTS waited for the whole clip** -> 100 ms MCP chunks **-> TTFB 1-7 s -> 0.84 s**
3 **Iris heard herself** -> WebRTC AEC + echo text filter **-> echo -57 dB (offline)**
4 **Music: turn never ended** -> music gate fast path **-> music stops in 0.1 s**
5 **Open: answers TV / side talk (2/2)** -> next: addressee check

refs: modelcontextprotocol.io · docs.pipecat.ai · fdbench · repo + logs: footer link